

## Case 36: Slope-factor Rothermel sweep

### 1 Verification purpose

This case verifies the scalar Rothermel slope response for FBFM2, FBFM9, GR2, TU5, and TL8 over 0, 10, 20, 30, 50, 75, and 100% rise over run. It tests slope-raster units, conversion to  $\tan^2 \phi$ , fuel-dependent packing response, and ELMFIRE's slope-factor cap without wind. The verification question is whether all 35 simulation conditions reproduce independently calculated intensity and slope-enhanced rate of spread (ROS) within the stated tolerances.

### 2 Mathematical formulation and numerical implementation

The no-wind response is

$$R = R_0(1 + \phi_s), \quad \phi_s = 5.275 \beta^{-0.3} \tan^2 \phi. \quad (1)$$

Here reported slope percent is rise/run times 100 and the raster value is the angle  $\phi = \tan^{-1}(s/100)$  in degrees. The formulation follows Andrews (2018), RMRS-GTR-371. The inspected ELMFIRE source additionally caps  $\phi_s$  at the maximum wind factor derived from its wind limit; the reference calculation reproduces and reports that implementation policy.

### 3 Derivation of expected behavior

Before the cap, plotting  $\phi_s$  against  $\tan \phi$  on logarithmic axes has slope two. For each fuel and raster angle, the local reference calculation calculates  $R_0$ , the uncapped factor, the cap, and final  $R$ . Because the initial level set is planar and aligned with the upslope direction, the output normal speed equals this scalar head-spread prediction.

The slope-only reference uses

$$\phi_s = 5.275 \beta^{-0.3} \tan^2 \theta, \quad R = R_0(1 + \phi_s).$$

As an explicit generated substitution, the requested 20 percent slope is stored as the integer raster angle  $\theta = 11$  degrees ( $100 \tan 11^\circ = 19.4380$  percent). For Fuel Model 2,  $\beta = 0.00575$ ,  $R_0 = 0.8237955 \text{ m min}^{-1}$ , and

$$\begin{aligned} \phi_s &= 5.275(0.00575)^{-0.3} \tan^2(11^\circ) = 0.936756, \\ R_{\text{ref}} &= 0.8237955(1 + 0.936756) = 1.595491 \text{ m min}^{-1}. \end{aligned}$$

Reaction intensity is unchanged by slope; here it is  $675.184 \text{ kW m}^{-2}$ . Every other generated fuel/slope pair uses the same equations with its stored integer angle and fuel packing ratio. Simulation condition errors are  $e_R = |R_{\text{max}} - R_{\text{ref}}| / \max(|R_{\text{ref}}|, 10^{-8})$  and analogously  $e_I$ . From  $t = a + bx_{\text{surface}}$ ,  $R_{\text{TOA}} = 60/b$  and  $R^2 = 1 - SSE/SST$ . The aggregate substitutions are  $\max e_R \leq 0.02$ ,  $\max e_I \leq 0.01$ ,  $\max e_{\text{TOA}} \leq 0.05$ , and  $\min R^2 \geq 0.995$ .

### 4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for direct ROS, 0.01 for reaction intensity, and 0.05 for time of arrival (TOA)-derived ROS; TOA regression requires  $R^2 \geq 0.995$ . The direct and reference calculation ROS

values are slope-surface speeds. Accordingly, the TOA regression converts map easting to distance along the sloping surface using  $x_s = x_0 + (x - x_0)/\cos\theta$  before fitting speed. Pointwise analytical errors are used instead of a trend-only test because degrees-versus-percent mistakes can preserve monotonicity. All simulation conditions and output fields must exist.

## 5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE’s supported  $[-1, 1]$  interval, preserving both an ignited interior and a resolved front gradient. For every positive reference spread rate, the prescribed and maximum timesteps use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured Courant–Friedrichs–Lewy (CFL) limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE’s stalled-front branch and corrupt the terminal timestamp. A zero-ROS simulation condition retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required simulation condition must reach the prescribed stop time in its generated namelist; a direct-only simulation condition may terminate early because lack of propagation is part of the state being tested. Each 1400 m square domain uses 10 m cells and a finite ignition strip with a planar east face in the interior analysis corridor. The strip does not touch the lateral grid boundaries. Wind is zero, dead moisture is fixed at 5/7/9%, and live moisture is 90%. Aspect is chosen so upslope is aligned with propagation. Slope inputs are generated as explicit integer degrees after applying the positive-value equivalent of Fortran NINT; the expected response and slope-surface coordinate conversion use that same generated angle. ROS should rise quadratically in  $\tan\phi$  before any source-defined plateau.

| Parameter     | Configured value                                    | Units            | Role                            |
|---------------|-----------------------------------------------------|------------------|---------------------------------|
| Domain/grid   | 1400 m square; 10                                   | m/cell           | Planar upslope propagation      |
| Fuel sweep    | FBFM2, FBFM9, GR2, TU5, and TL8                     | code/name        | Fuel-dependent packing response |
| Slope sweep   | 0, 10, 20, 30, 50, 75, and 100%                     | percent rise/run | Independent terrain target      |
| Wind/moisture | calm; dead 5/7/9, live 90/90                        | mph / percent    | Isolates slope factor           |
| Timing        | generated whole-second step and exact-multiple stop | s                | CFL-safe terminal evidence      |

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

Whole-domain input configuration — variants/fuel\_102\_s\_030; WS=0 mph, WD=270 deg

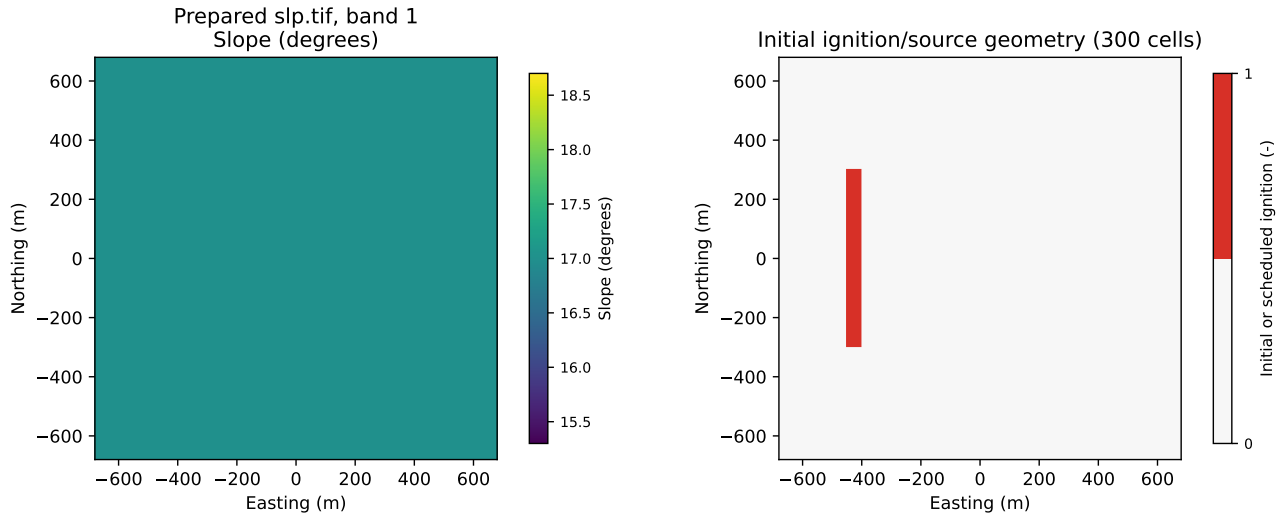


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from  $\phi \leq 0$ .

## 6 Actual simulation results

The available results cover 35 of 35 simulation conditions and has state **PASS**.

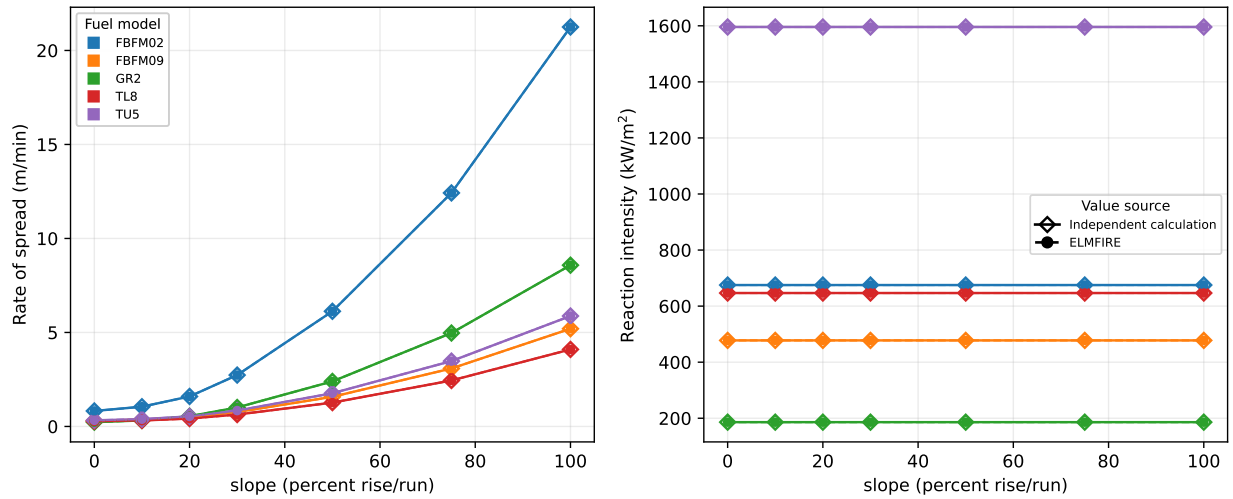


Figure 2: Independent calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines). Each fuel-model pair uses the same color.

## 7 Calculated metrics and verification decision

| Metric                                    | Limit        | Calculated             | Status |
|-------------------------------------------|--------------|------------------------|--------|
| maximum direct-ROS relative error         | $\leq 0.02$  | 5.13349e-07 fraction   | PASS   |
| maximum reaction-intensity relative error | $\leq 0.01$  | 2.93279e-07 fraction   | PASS   |
| maximum TOA-ROS relative error            | $\leq 0.05$  | 0.0265981 fraction     | PASS   |
| minimum TOA regression R2                 | $\geq 0.995$ | 0.999961 dimensionless | PASS   |

The overall decision is **PASS**.

## 8 Technical interpretation

Large errors growing with slope indicate a units or tangent error. Fuel-dependent errors implicate packing ratio or table selection. A premature or absent plateau indicates changed cap semantics. An accurate direct field but biased TOA indicates propagation or coordinate handling.

## 9 Scope and limitations

This test covers uniform positive slopes aligned with the front. It does not verify vector competition between wind and slope, complex terrain, or experimental fire behavior; CASE40 covers the directional coupling.